Coastal Education & Research Foundation, Inc.

Distribution of Phosphorus and Iron Species in the Sediments along the Path of Yellow River and Correlation of Contents of the Two Species

Author(s): Weihua Lv and Hongwei Yang

Source: *Journal of Coastal Research*, FALL 2020, SPECIAL ISSUE NO. 104. Advances in Water Resources, Coastal Management, and Marine Science Technology (FALL 2020), pp. 813-817

Published by: Coastal Education & Research Foundation, Inc.

Stable URL: https://www.jstor.org/stable/10.2307/48641011

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms



Coastal Education & Research Foundation, Inc. is collaborating with JSTOR to digitize, preserve and extend access to *Journal of Coastal Research*

This content downloaded from 147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms

Journal of Coastal Research	SI	104	813–817	Coconut Creek, Florida	2020

Distribution of Phosphorus and Iron Species in the Sediments along the Path of Yellow River and Correlation of Contents of the Two Species

Weihua Lv[†] and Hongwei Yang[†‡*]

[†]Chemistry and Environment Science College
Inner Mongolia Normal University
Hohhot 010022, China

[‡]Key Laboratory of Environmental Chemistry
Inner Mongolia Normal University
Hohhot 010022, China

www.cerf-jcr.org

www.JCRonline.org

ABSTRACT

Lv, W.H. and Yang, H.W., 2020. Distribution of phosphorus and iron species in the sediments along the path of yellow river and correlation of contents of the two species. *In:* Guido Aldana, P.A. and Kantamaneni, K. (eds.), *Advances in Water Resources, Coastal Management, and Marine Science Technology*. *Journal of Coastal Research*, Special Issue No. 104, pp. 813–817. Coconut Creek (Florida), ISSN 0749-0208.

We measured contents of total phosphorus (TP) and total iron (TFe) in the sediments from 21 samples along the path of Yellow river. TP content of the 21 sediments were from 0.87 to1.75 g·kg-1, and TP contents of sediments from Huayuankou section (H18) and Liulin section (H13) were maximum and minimum, respectively. TFe content were in the range of 15.50~31.24 g·kg-1, correspondingly, Liulin section (H13) and Sanmenxia section (H17), were the maximum and minimum. It had a positive correlation relationship between TFe and TP by correlation analysis on Maduo section (H1), Xinghai section (H2), Zhongwei section (H5), Baotou section (H10), Xingxian section (H12), Dayudu section (H15), Sanmenxia section (H17), Jinan section (H19) and Yellow River Mouth section (H20), and the equation fits well (R2>0.93). The distribution of TP and TFe of the sediments from different sections in Yellow River related with regional environmental conditions and the geological environment.

ADDITIONAL INDEX WORDS: *Yellow River, sediments, TP and TFe, correlation, distribution characteristics.*

INTRODUCTION

Phosphorus is an significant biogenic element, and also the major restriction of water body eutrophication. Its migration and circulation in the sea-land interaction directly affect the primary productivity of the water body (Almasri, Saleh, and Atieh, 2019; Cui, Ou, and Wang, 2019; Karczmarczyk and Renman, 2011; Rodriguez *et al.*, 2018; Tang ,Wu, and Li, 2018). Trace element iron is necessary for people, as well as a limiting factor for the growth of planktonic algae. And iron has a strong adsorption due to its oxides surface characteristics (Zou, Lue, and Jiang, 2009). Therefore, the geochemical cycle of iron can affect the cycle and transformation of phosphorus and other nutrients in the soil. Studies (Deng *et al.*, 2019; Huang *et al.*, 2018; Krom *et al*., 1991; Peng, Tian, and Yang, 2019; Rahav *et al*., 2013; Ralph *et al.*, 2018; Zou, Lue, and Jiang, 2009) have indicated that phosphorus and irone as an significant indicator can be used to evaluate pollution degree, and releasing of phosphorus in sediments serious affecting the nutritional status of water bodies, in addition, the total phosphorus in sediments can combine with iron and other minerals by physical, chemical and biological interaction. So it is important for further analysis of their pollution trends by determining the content of TP and TFe in sediments and analyze their correlation.

DOI: 10.2112/JCR-SI104-139.1 received 9 December 2019; accepted in revision 12 May 2020.
*Corresponding author: nsdyhw@163.com
©Coastal Education and Research Foundation, Inc. 2020

Yellow River is a less water and more sand river in the world, and phosphorus load in the sediments is higher, and total phosphorus amount (1.36 g·kg-1) is larger than that of some serious eutrophication lakes , for example, Taihu Lake (0.88 g·kg-1) (Jin, Meng, and Jiang, 2006; Zhang *et al*., 2009), Chaohu Lake (0.55 g·kg-1) (Wang *et al*., 2007) *etc*. In recent years, with the rapid economic development, Yellow River has a tendency to exceed its own carrying capacity because of discharging industrial, agricultural waste water and domestic sewage. So it attracts much attention on the ecological environment investigation of Yellow River recently (Liu, Li, and Dong, 2012; Sun, 2006; Zhang *et al*., 2008; Zhang, Pan, and Chen, 2009), but most research are concentrated on the iron or phosphorus forms of Yellow River sediments separately (Chen *et al*., 2019; Peng *et al*., 2019), yet few studies on the effect of total iron on total phosphorus. For the purpose, we select Yellow River as investigate subject and study the total iron and total phosphorus distribution characteristics from the perspective of the whole basin, analyze their correlation, to explore the effect of iron on the release of phosphorus.

MATERIALS AND METHODS

21 sampling points were set up along the main Yellow River to collect samples, with sample numbers of H1 ~ H21 respectively, and Figure 1 showed the sampling point locations. All the samples were collected using GPS during the period from 2017 to 2018. The samples were drained and then put into a plastic sampling bag. Air of the plastic sampling bag was discharged and the bag was sealed. The samples were delivered to the laboratory and

This content downloaded from
147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms

cryopreserved. When used, a proper amount of sediment was taken out and dried naturally, and stored at 4°C for later use.

Instruments and Reagents

Anke-40BC entrifuge, HY-2 multispeed oscillator, 722-N type spectrophotometer. KH_2PO_4, $K(SbO)C_4H_4O_6 \cdot 1/2H_2O$, $K_2S_2O_8$, $HClO_4$, HF, phenanthroline, ammonium molybdate, ascorbic acid were used in this study. Water used in present work was double-distilled water.

Determine Content of TP

Samples were digested by $K_2S_2O_8$. 0.1 g of sample was precisely weighed and 10 mL secondary water was added to the sample, then 3mL $K_2S_2O_8$ aqueous and 37 mL secondary water were added to the sample again. After that, sample was moved into the pressure cooker and was presurrized for half hour and then was centrifuged. TP content was measured after filtered with 0.45 μm membrane using Mo-Sb-Anti-Sepetrochrometry. 20 mL sample was moved into 25 mL colorimetric tube. 2.5mL ammonium molybdate and 0.5 mL ascorbic acid were added to the sample and then made 25 mL. The sample was heated in water bath at 30 °C for 30 minutes, and then absorbance was measured at 710 nm.

Determine Content of TFe

Samples were digested with HF-$HClO_4$-HNO_3. The detailed procedure was as follows: 0.2000 g sample was weighed and moved into a small beaker (50 mL), 5 mL concentrated HNO_3, 1 mL $HClO_4$ were added to electric hot plate and decomposed under low temperature. 10 mL HF, 1 mL $HClO_4$ were added to the sample and digested under high-temperature, and the sample was concentrated by evaporation to dry 5 mL of 6 mol·L^{-1} with dilute hydrochloric acid was added to the sample and the sample was made 50 mL in a flask.

Iron was measured with phenanthroline spectrophotometry: taking 10 mL sample solution, and l mL of 10% Hydroxylamine Hydrochloride was added in, 2 minutes later, 5 mL of 1 mol • L^{-1} sodium acetate and 2 mL of 0.1% phenanthroline then were added. Adjust pH to 5-7 after 5 minutes. At last, absorbance was determined at 510 nm after making the volume up to 25 mL by secondary water.

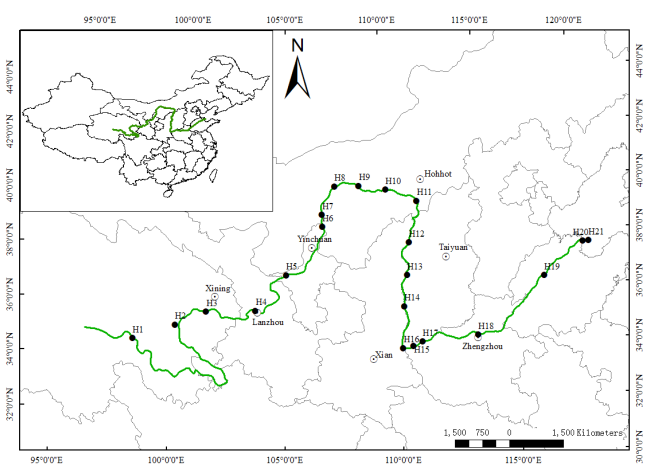


Figure 1. Sample stations along the path of Yellow River.

RESULTS AND DISCUSSION

Characteristics of TP Distribution in the Sediments

TP and TFe content in the sediments of Yellow River were indicated in Table 1 and Figure 2.

TP content of 21 samples was from 0.87 to 1.75g·kg^{-1}, with the ascending order of H13<H14<H8<H3<H11<H4<H16<H17<H2<H1<H7<H20<H12<H15<H10<H6<H21<H9<H19<H5<H18. Minimum and maximum values came from Liulin and Huayuankou section. Liulin section is located in the east bank of the middle reaches of Yellow River, and belongs to the northwest loess plateau and loess hilly and gully region. The sediment mineral components of Yellow River in Liulin section are mainly quartz and feldspar, and the clay minerals are mainly illite and montmorillonite. Peng, Tian, and Yang (2019) investigated phosphorus forms and the results displayed that the content of calcium-bound phosphorous (P_{Ca}) of samples from Liulin section was the minimum of the 21 samples and only 0.75 g·kg^{-1} which lead to a low content of TP. P_{Ca} was not only the main form of phosphorus in the s ediments and but also the relatively inert phosphorus component, thought to be bio-unavailable. It was formed naturally or by rock debris, depending on the geological Conditions.

P_{Ca} of sample from Huayuankou section was maximum and as high as 1.47 g·kg^{-1}, lead to a larger TP content. In addition, Huayuankou was an important historical city in the Central Plains of China, where Yellow River warping irrigation area was established, survived and developed, and is planned as Huayuankou Irrigation Area. After the completion of Huayuankou Irrigation District, the development of agricultural production in Zhengzhou has been greatly improved. At the same time, the new problem was serious, leading local farmers to withdraw irrigation and switch to well irrigation. As a result, some cultivated land was severely salinized. Fertilizer was an significant factor on migration of phosphorus. Organic fertilizers and organic phosphorus pesticides in soil released, then entered the water. It could result in rising of organic phosphorus concentrations (Sun, Qiu, and Gao, 2019). And this may be why there hais a larger TP content in Huayuankou.

Table 1. *TP and TFe content in sediments of the 21 samples.*

Sample	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11
TFe (g·kg^{-1})	27.58	28.48	24.45	22.41	28.59	28.64	26.75	17.97	24.20	27.36	24.51
TP (g·kg^{-1})	1.42	1.41	1.21	1.32	1.72	1.66	1.49	1.18	1.69	1.62	1.31
Sample	H12	H13	H14	H15	H16	H17	H18	H19	H20	H21	
TFe (g·kg^{-1})	23.09	15.50	17.83	28.94	22.90	31.24	27.14	26.31	23.06	27.84	
TP (g·kg^{-1})	1.55	0.84	0.94	1.56	1.37	1.39	1.75	1.71	1.49	1.68	

This content downloaded from
147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms

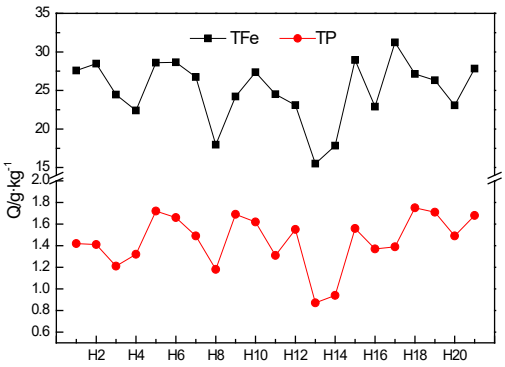


Figure 2. TP and TFe contents in sediments from 21 samples positions of Yellow River.

The TP content in sediment samples from Liulin (H13) and Hukou (H14) sections of Yellow River both were less than 1.00 g·kg^{-1}. Liulin and Hukou section have similar geographical locations and the same soil types, so the difference of TP content of the samples from the two sections is small. Content of TP of the samples from the other sections was from 1.20 to 1.75g·kg^{-1}, and distributed evenly, relatively, indicating that Yellow River had not been seriously polluted.

Characteristics of TFe Distribution of the Sediments

TFe contents of the sediments were 15.50~31.24 g·kg^{-1}. The TFe content increased in the order of H13< H14< H8< H4< H16< H20< H12< H9< H3< H11< H19< H7< H18< H10< H1< H21< H2< H5< H6< H15<H17, Minimum and maximum values corresponded to the samples from Liulin and Sanmenxia section, respectively. The content of TFe from Liulin section (H13), located in the middle reaches of Yellow River, was the lowest, only 15.50 g·kg^{-1}. The reason for the low TFe content is the geological and geomorphic features. Residual iron of the samples from Liulin was minimum as low as 13.77g·kg^{-1}, accounted for 88.84% of TFe. Residue-Fe combining with mineral elements mainly came from natural minerals. Residue iron was a bio-unavailable component. The low residual iron content is the reason for the low TFe content. TFe Content of Hukou section (H14) was 17.83 g·kg^{-1}. And Hukou section neighbours the Liulin section, both geological environment are similar. The residual iron content from Hukou section was as low as 14.76 g·kg^{-1} lead to the low TFe content. The TFe content of Sanmenxia section in the middle Yellow River (H17) was the maximum and as high as 31.24 g·kg^{-1}. There widely existed bauxite in Sanmenxia. Andand bauxite bump layer was iron weathering crust，thus came into being thick iron clay layer. Iron clays were disturbed by monsoon. Blown Sand carried large amounts of iron clay particles into Yellow River and accumulated in the sediments due to the disturbed of monsoon, it was the first reason why there has a larger TFe content in Sanmenxia sediments. Diaspore was one of the main minerals of bauxite ore, and the element Fe was widespread in the formation process of diaspore, which was the second reason for the high TFe content of Sanmenxia section. The construction of Sanmenxia Water Control Project on Yellow River was the third reason for the high TFe content of Sanmenxia section.

TFe content of sediments in Yellow River was within 15.55~31.24 g·kg^{-1}, significantly different existing with different areas was chiefly owing to the difference of geological environment, human activities and industrial and agricultural pollution.

Correlation between TP and TFe in the Sediments

Spss23.0 software was used to analysis the correlation between TP and TFe contents in sediments of Yellow River. The results showed that TP and TFe of samples H1, H2, H5, H10, H12, H15, H17, H19, H20 have a positive correlation relationship and the equation fit well (R^2>0.93). This indicates that TP in the above sediments can be indirectly measured by TFe content. Studies have shown that the phosphorus release capacity of the sediment is controlled by iron, that is, FeO(OH) has a strong adsorption capacity for inorganic phosphorus under aerobic conditions; FeO(OH) is reduced and dissolved, the adsorbed phosphorus was released under anaerobic conditions at the same time. Yellow River has a high pH value (between 8.00 and 8.50). Iron hydroxide in sediments dissolves and releases adsorbed phosphorus because of oxidation-reduce action at anaerobic conditions under such water environment. The coupling of iron reduction and oxidative degradation of organic matter is one of the most important biogeochemical reactions in the early diagenesis of sediments. The contents of TP and TFe in the sediments of the above 9 sections are positively correlated, which reflects the environmental conditions of various places to a certain extent. It can be seen that iron in the sediments is one of the key factors controlling the release of endogenous phosphorus, and plays an important role in regulating the ecological environment quality and biological productivity of water bodies.

TP and TFe of samples from the other sampling stations have a poor correlation results, perhaps due to that iron binding forms was complicated, and that TP and TFe may also correlated with other minerals, such as, organic matters, manganese oxides, etc. In addition, related studies have shown that the average content of iron in the sediments is also related to the sediment environment, and environmental factors play an important role in the accumulation of iron in the sediments. Therefore, the specific reasons for the influence of iron content in the sediments of other sampling sections need to be further studied.

Comparison with Other Water Bodies on TP and TFe

Table 2 indicated TP and TFe contents at the sediments in Yellow River approximated those in Taihu Lake, Xuanwu Lake, middle and lower reaches of Yangtze River. However, they were lower than Haihe and Donghu Lake. Tai Lake is an important natural lake in middle and lower reaches of Yangtze River. Xuanwu Lake was a representative city lake in Nanjing. Studies of Meng, Jiang, and Jin (2004) showed that illite/montmorillonite mixed layers were the mainly components of clay minerals of lake sediments in middle and lower reaches of Yangtze River. It is showed that in 10 sedimentsthe major components of clay minerals were illite/smectite mixed layers in Yellow River upper and middle reaches by Wang, Bao, and Guo (2009), which may be the reason why its TP and TFe contents approached other water bodies like Tai Lake, middle and lower reaches of the Yangtze River, Xuanwu Lake. Wuhan East Lake pollution is increasing, and a large amounts of nutrient salts were deposited in the lake body of East Lake each year (Meng, Yao, and Yu, 2006). Haihe is one of the major routes

This content downloaded from
147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms

Table 2. *TP and TFe content in different water bodies sediments.*

Water Bodies	TFe Content/ μm^{ol}·g-1	TP Content/ μmol·g^{-1}
Tai Lake	231~688	7.1~80.6
Middle and lower yangtze		9.92~46.92
East Lake		22.2~35.81
Xuanwu Lake		8.19~90.6
Thames Piver	12~8.333	1.7~649
Haihe	595.67~719.91	29.00~78.99
Yellow River	276.79~557.86	9.18~18.43

from Bohai Bay to sea, and which was seriously polluted, so these maybe result in larger TP and TFe contents in Haihe River and East Lake sediments than Yellow River.

CONCLUSIONS

TP content was 0.87~1.75 g·kg^{-1} in sediments of Yellow River, and that of Liulin and Huayuankou sections of Yellow River were the minimum and maximum, respectively. TFe content was 15.50~31.24 g·kg-1, and TFe of samples from Liulin and Sanmenxia sections were minimum and maximum, respectively.

There was a positive correlation relationship between TP and TFe of samples H5, H10, H12, H15, H17, H19, H20, and the equation fit well (R^2>0.93), indicating that TP of sediments at these sections could be indirectly evaluated by TFe, there had poor correlation relationships between TP and TFe in other sediments .

Comparison showed that the contents of TP and TFe of sediments of Yellow River were close to those of Taihu Lake, Xuanwu Lake and the middle and lower reaches of the Yangtze River, but lower than those of Haihe and Donghu Lake.

ACKNOWLEDGMENTS

Funding by Inner Mongolia Normal University natural science research foundation(2016ZRYB010).

LITREATURE CITED

Almasri, D.A.; Saleh, N.B., and Atieh, M.A., 2019. Adsorption of phosphate on iron oxide doped halloysite nanotubes. *Scientific Reports*, 9, 3232.

Chen, Y.; Lin, H.; Shen, N.; Yan, W.; Wang, J.A., and Wang, G.X., 2019. Phosphorus release and recovery from Fe-enhanced primary sedimentation sludge via alkaline fermentation. *Bioresource Technology*, 278, 266-271.

Cui, H.; Ou, Y., and Wang, L.X., 2019. Distribution and release of phosphorus fractions associated with soil aggregate structure in restored wetlands. *Chemosphere*, 223, 319-329.

Deng, Y.X.; Zhang, T.; Sharma, B.K., and Nie, H.Y., 2019. Optimization and mechanism studies on cell disruption and phosphorus recovery from microalgae with magnesium modified hydrochar in assisted hydrothermal system. *Science of the Total Environment*, 646, 1140-1154.

Huang, X.D.; Zhang, Q.Y., and Ma, X.L., 2018. The thermal comfort models of venue in hot and humid subtropical regions. *Ccamlr Science*, 25(3), 244-252.

Jin, X.C.; Meng, F.D., and Jiang, X., 2006. Physical-chemical characteristics and of phosphorus speciations in sediments of northeast lake Taihu. *Resources and Environment in the Yangtze Basin,* 15(3), 388-394.

Karczmarczyk, A. and Renman, G., 2011. Phosphorus accumulation pattern in a subsurface constructed wetland treating residential wastewater. *Water*, 3, 146-156.

Krom, M.D.; Kress, N.; Brenner, S., and Gordon, L.I., 1991. Phosphorus limitation of primary productivity in the Eastern Mediterranean-Sea. *Limnology and Oceanography*, 36, 424-432.

Liu, F.; Li, X.Q., and Dong, G.C., 2012. Nutrients distribution and eutrophication assessment on coastal waters adjacent to Yellow River delta. *Journal of Hydroecology*, 33(1), 15-19.

Meng, F.; Jiang, X., and Jin, X.C., 2004. Physical-chemical characteristics of the sediments in lakes from the middle and lower reaches of the Yangtze River. *In:* Liu H.L. (ed.), *Environmental Science and Resource Utilization*. *Research of Environmental Science*, Special Issue No. 1, pp.24-29.

Meng, J.; Yao, P., and Yu, Z.G., 2006. Speciation, bioavailability and preservation of phosphorus in surface sediments of the Changjiang Estuary and adjacent East China Sea inner shelf. *Environmental Pollution*, 139, 288-295.

Peng, Y.H.; Tian, C.C., and Yang, H.W., 2019. Distribution of phosphorus species and their release risks in the surface sediments from different reaches along Yellow River. *Environmental Science and Pollution Research*, 26, 28202-28209.

Rahav, E.; Herut, B.; Stambler, N.; Mulholland, M.R., and Berman-Frank, I., 2013. Uncoupling between din itrogen fixation and primary productivity in the eastern Mediterranean Sea. *Journal of Geophysical Research-Biogeosciences*, 118, 195-202.

Ralph, P.; Maxwell, N., and Silaban, D., 2018. The morphology features of culm and vascular bundles of bamboo gigantochloa apus. *Ccamlr Science*, 25(3), 197-201.

Rodriguez, C.; Iglesias, K.; Bícego, M.C.; Taniguchi, S.; Sasaki, S.T.; Kandratavicius, N.; Bueno, C.; Brugnoli, E., and Venturini, N., 2018. Hydrocarbons in soil and meltwater stream sediments near Artigas Antarctic Research Station: Origin, sources and levels. *Antarctic Science*, 30, 170-182.

Sun, Y.M.; Qiu, T.L., and Gao, M., 2019. Inorganic and organic fertilizers application enhanced antibiotic resistome in greenhouse soils growing vegetables. *Ecotoxicology and Environmental Safety*, 179, 24-30.

Sun, Z., 2006. Study on eutrophication advancement for the Yellow River estuary reservoirs. Harbin: Harbin Institute of Technology.

Tang, X.Q.; Wu, M., and Li, R., 2018. Distribution, sedimentation, and bioavailability of particulate phosphorus in the mainstream of the Three Gorges Reservoir. *Water Research*, 140, 44-55.

This content downloaded from
147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms

Wang, X.L.; Bao, H.Y., and Guo, B.S., 2009. Physical-chemical characteristics and phosphorus speciation of the sediments in the upper and middle reaches of the Yellow River. *Environmental Science*, 30, 720-725.

Wang, X.W.; Wang, X.Y.; Feng, Y., and Xue, J.P., 2007. Study on content of total phosphorus and forms of inorganic phosphorus in sediments of chaohu Lake. *Journal of Soil and Water Conservation,* 21(4), 56-59.

Zhang, J.M.; Liu, S.; Zhang, Q., and Liu, Y.T., 2008. Nutrient distribution and eutrophication assessment for the adjacent waters of the Yellow River Estuary. *Marine Science Bulletin*, 17(5), 65-72.

Zhang, X.W.; Pan, G., and Chen, H., 2009. Phosphorus forms and distribution in the sediments from different reaches along the Yellow River. *Acta Scientiae Circumstantiae,* 29(1), 191-198.

Zou, Y.C.; Lue, X.G., and Jiang, M., 2009. Distribution of sediment iron of the ditch system in Sanjiang Plain, northeast China. *Environmental Science*, 30, 889-893.

This content downloaded from
147.147.41.114 on Tue, 16 Jun 2026 12:30:48 UTC
All use subject to https://about.jstor.org/terms